

Review-II — what to say, slide by slide

GaN Based Synchronous Buck Converter with an Improved Gate Driver

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Guide: Dr. Bindu — SENSE, VIT Chennai · Review-II, 30.09.2026

26 slides. **Total 14 min 25 s of speaking.** Bracketed times are cumulative:

if the clock is past one, you are behind. Everything in **bold** is a number

you commit to out loud — each one is on the slide behind you, so you are

never quoting something the examiner cannot see.

Speaker split — swap it if you like, but decide before you walk in, and

whoever is not speaking watches the clock and the laptop.

Slides	Who	What they own
1–9	Amritha	the problem, the aim, what we built and why
10–18	Sanjay	implementation, method, the two runs
19–26	Aamir	the demonstration, the comparison, what is left

The three slides that decide the review are 9, 21 and 23. If you are

running out of time, cut 8 and 12. Never cut those three.

Do not read the captions aloud. They are there so the examiner can check

you afterwards. Say the sentences below and point at the picture.

1 · Title — 20 s (0:20)

Good morning. Our project is a **GaN** based synchronous buck converter, and the part we designed is the gate driver inside it.

A buck converter takes a DC voltage and gives a lower DC voltage. Ours takes **100 V** in and gives **48.50 V** out.

Get the signed slide on screen first. Do not read the registration numbers.

2 · Problem Statement & Background — 60 s (1:20)

Start from the device, not from the literature.

A half-bridge is two transistors in series across the supply. The rule is simple: exactly one of them is on at a time. If both are ever on together,

the supply is shorted through them.

GaN transistors are used because they switch very fast. That is the whole reason to choose GaN — and it is also what causes our problem.

When the bottom device turns on, the voltage at the midpoint collapses in about three nanoseconds. There is a small capacitance inside the top device, between its drain and its gate. A fast voltage change across a capacitor pushes a current through it, and that current goes straight into the gate that is supposed to be held off.

We measured that gate. It rises to **1.65 V**. The device turns on at **1.4 V**. So the device that should be off turns on. That is the fault.

Solutions exist in the literature — drive in steps, clamp the gate, adjust the dead time, hold the gate negative. They report large gains. But they report them as one number.

Our gap is this. Those gains mix two different things: picking one good setting once, and changing the setting while the converter is running. Only the second one needs a sensor, a lookup table and a controller. Nobody separates them, so nobody knows whether that hardware is worth building.

3 · Aim and Approach — 45 s (2:05)

So our aim is one question: how much of a gate driver's benefit actually needs the driver to re-tune itself while the converter runs, and how much you get from choosing one good setting and leaving it alone.

To answer it we needed a driver whose every setting we could change and measure. That is the segmented driver: eight pull-up steps, eight pull-down steps, an adjustable dead time, a Miller clamp, and an off rail we can take to minus two volts.

Four steps. Build the converter and check it actually converts. Recreate the fault. Fix it one change at a time, so each fix is measured on its own

run. Then ask whether one fixed setting is enough.

4 · System Architecture — 40 s (2:45)

Point along the arrows, left to right. Do not read the boxes.

One signal path. A PWM command comes in. The FPGA controller holds the control word — that word is what sets the driver's strength. The segmented driver shapes the switching edge. The half-bridge drives the load.

The block in green is what this project designs. It is drawn once and placed twice, once for each gate.

Two things to notice. The control word is set at power-up and never changed — there is no sensor and no lookup table in this path. And the red arrow is the fault: the switching edge coupling back into the gate that should stay off.

5 · Converter Under Simulation — 30 s (3:15)

This is the converter itself, and it is drawn from the netlist we simulate, so every value on this sheet is the value that ran.

A hundred volts in on the left, the two GaN devices in the middle, the filter and a ten ohm load on the right. It delivers **48.50 V** at **236.26 W**, **97.49 %** efficient. The converter works before we start arguing about the driver.

6 · Novelty: Three Added Blocks — 50 s (4:05)

This is the slide that answers "what is actually yours". Be precise, and be honest about what is not yours.

The base paper is Zhang and others, ISPSD 2020. This table is their design against ours, and only the shaded rows differ.

The command is theirs. The power stage is theirs. The idea of segmenting the output stage is theirs — we are not claiming that.

Three blocks are added. One: the drive strength is a digital control word written by the FPGA, where they use one analogue bias resistor fixed at fabrication. Two: an always-on active Miller clamp, which they do not have. Three: an off rail selectable to minus two volts, where theirs is tied to zero.

The last row is what those three are worth at our headline operating point. Their crosstalk margin is **0.407 V**. Ours is **2.576 V**. From one fixed setting.

7 · Novelty at Circuit Level — 40 s (4:45)

Same claim, now at circuit level. The converter is on top with the two gates ringed — that is what a gate driver drives. Underneath is what sits behind those gates, theirs on the left and ours on the right.

Both of our additions are established practice, and we cite the papers. What is new is not the idea of a clamp or a negative rail. It is that the base paper has neither, and that we measure what each one is worth on this converter.

The clamp alone is worth **0.82 V** of margin. The rail alone is worth **2.01 V**. Together, minus **0.249 V** becomes **2.576 V**.

8 · Gate Driver Schematics in KiCad — 20 s (5:05)

Cut this slide if you are behind. One sentence otherwise.

The same two sheets, whole, at the same scale, so you can see nothing has been cropped out. These are screen captures of KiCad, not redrawings.

9 · Crosstalk Mechanism at Device Level — 55 s (6:00)

This is the slide that shows you understand the physics. Slow down.

On the left is the mechanism. The bottom device turns on, the switch node falls from **106 V**, and at its steepest that edge is **124 V/ns**. That rate of change, times the drain-gate capacitance, is a current — and it has nowhere to go except into the gate of the top device.

On the right is what the driver does about it. The gate node has three paths to hold it down. The pull-down slices — the more you turn on, the harder the gate is held. The Miller clamp — its own switch and a half-ohm path, opened exactly when the other device switches. And the off rail, which sets where the gate starts from.

The important point is that the clamp and the rail are not the same fix. The rail moves where the gate starts. The clamp shortens how far it gets lifted. Two different mechanisms, so they add instead of overlapping. That is why the numbers on the last slide added up.

10–12 · Implementation, three sheets — 35 s total (6:35)

Sanjay takes over. One sentence each, do not dwell.

(10) This is the converter as an actual KiCad sheet, drawn from the netlist, with the two gate drivers as sub-sheets of this page.

(11) This is our driver. Eight pull-up slices, eight pull-down, the Miller clamp on the right, the off rail selectable to minus two volts.

(12) And this is the base paper's driver, rebuilt by us in the same testbench: seven slices per bank in two stages, one bias resistor, no clamp, off rail tied to reference.

13 · Segmented Driver: Block Structure — 35 s (7:10)

Here is how a slice works, because the whole study turns on it.

Each slice is one switch and one resistor. The control word decides whether that resistor is eight ohms and in circuit, or one gigaohm and effectively out. So drive strength is just how many of the eight parallel paths are live — that is all "segmented" means.

On the right, the clamp: its own switch, half an ohm to the bottom rail, and that rail itself selectable to minus two volts.

14–15 · The two files we wrote — 30 s total (7:40)

(14) This is the actual SPICE source of that output stage, at its real line numbers. The clamp is deliberately not one of the slices — it has its own switch and its own timing, because it has to hold the gate down while the *other* device is switching.

(15) And this is the GaN device model. Vendor models are LTspice-only and do not port, so this is written from the EPC2010C datasheet. One consequence matters: GaN has no body diode, so reverse conduction costs extra, and that is exactly what the minus two volt rail has to pay for in dead time. We are not hiding that — it is the cost side of our own result.

16 · Methodology — 40 s (8:20)

If you are asked one methodology question, it is this slide.

Every comparison in this deck is the same file with one thing changed.

Held identical: the netlist, byte for byte — not a copy, the same file. The device model on both sides. The parasitics. The operating point. The solver settings.

The one thing that moves is named in the right-hand column. For theirs against ours, only the driver subcircuit is swapped.

And we give their driver an advantage it does not have in its own paper: we search both of its controls at every operating point and run it at whatever

setting wins. Ours runs one fixed word everywhere. So any difference left is the driver, and it is the comparison that flatters them.

17–18 · The two runs — 45 s total (9:05)

These slides carry video. Click, let it run, talk over it.

(17) Their driver, in our testbench. The gate that should stay off peaks at **0.993 V** against a **1.400 V** threshold. That is **0.407 V** of margin — it does not fail, but that is what it has left.

(18) The same sequence, our driver. The gate peaks at minus **1.176 V** — **2.576 V** of margin, six times theirs. Same axes, fixed in the script, so the two pictures are directly comparable. And this is one fixed control word, the same one used at every corner in the study.

19 · Demonstration — 50 s (9:55)

Aamir takes over. Play the film and narrate only the parts it is showing.

This is the whole thing end to end. The circuit as its KiCad sheet, the tools named and versioned, then their driver and ours on the same bench at a hundred volts and ten amps.

Theirs peaks **0.993 V**, short of the threshold. Ours peaks minus **1.176 V**, well clear. And the converter it drives is regulating at **48.50 V**, **236.26 W**, **97.49 %** efficient.

Every number on that screen was computed while the film was being built, not typed into a caption.

20 · Live Simulation — 70 s (11:05)

This is the slide she asked for. Run it live. Terminal ready beforehand, in the repository, font already large.

I will run it now rather than show you a recording.

Type and run:

```
bash proof/LIVE-SIM.sh
```

While it runs — it takes about ten seconds:

It is running two transients of one circuit file. Between the two runs, exactly two things change: the Miller clamp is enabled, and the gate off rail is moved. Nothing else.

When the numbers print:

There it is. It prints what it just measured next to what our slides claim. The off gate reaches **1.649 V** against the **1.65** on our slide, and the margin is **2.576 V** against the **2.576** on our slide. Then it draws the two waveforms those runs produced.

If it fails to run, say this and move on — do not debug in front of her:

The recorded run of the same command is on the slide behind me, and the script is in the repository.

If she asks to see the converter itself running, not just the gate —

there is a second command, about twenty-three seconds:

```
bash proof/LIVE-BUCK.sh
```

This one runs the whole converter rather than the gate. It prints the output voltage, the power and the efficiency it just measured beside what our slides claim — **48.50 V**, **236.26 W**, **97.49 %** — and it also prices the fix: the crosstalk fix costs about a quarter of an efficiency point.

Do not run both unprompted. Slide 20 is the one you owe her; this is the answer to a follow-up.

21 · Crosstalk: Fault and Mitigation — 55 s (12:00)

The result slide. Point at the four panels in order.

Two runs, four panels. Top row is the cause, bottom row is the effect.

Top: the switch node falls, and at its steepest it is **124 V/ns**. That is the aggressor, and it is identical in both runs.

Bottom left, the gate with no clamp and a zero volt off rail. It starts at about zero, gets lifted **1.643 V**, and peaks at **1.649 V** — past the **1.4 V** threshold. False turn-on.

Bottom right, same run with the clamp on and the rail at minus two volts. It starts at minus **1.941 V**, is lifted only **0.765 V**, and peaks at minus **1.176 V**. No false turn-on.

And you can read both fixes off this picture separately. The rail moved where the gate starts. The clamp cut the lift from **1.643 V** to **0.765 V**. That is the two mechanisms, measured separately, on one slide.

22 · Six-Parameter Comparison — 30 s (12:30)

Silicon, then the base paper, then ours — same converter, same device class.

Latency goes **17.55** nanoseconds, to **4.04**, to **2.78**. Device power goes **8.60 W**, to **2.93**, to **2.60**.

Be careful how you read that: the device swap accounts for most of it and the control swap for the rest. We are not claiming the whole ratio.

Overshoot is the one row that runs the other way, and it runs that way for the same reason the other five do not: silicon does not overshoot because its edge is nine times slower — and that slowness is exactly what costs it six watts. Speed and device stress are the same knob.

23 · Comparison with the Base Paper — 70 s (13:40)

The slide the review turns on. Do not rush the last paragraph.

Four operating corners, mildest to hottest. Same netlist, same devices, only the driver swapped.

Their driver as their paper builds it, then their driver re-tuned at every corner — which is more freedom than their design actually has, because their scheme sets one bias resistor once. Then ours, at one fixed word.

Two things matter here, and neither is the ratio.

First, the gap widens under stress. Their best margin falls from **0.503 V** at the mildest corner to **0.181 V** at the hottest. Ours goes **2.757 V** to **2.251 V**. They degrade exactly where it matters, because a clamp does not care how hot the device is. The worst corner is what a converter has to survive, and that is **0.181 V** against **2.251 V**.

Second — and this is the part we would defend hardest — their scheme buys its margin by slowing the edge down, sixty-seven to a hundred and three volts per nanosecond. Ours runs a hundred and one to a hundred and seventy-five — about twice as fast — and still wins by a factor of several. The margin is not paid for with switching speed. That is the useful form of the result.

And what it costs us. Our turn-on energy is higher at three of the four corners, because the negative rail deepens GaN's reverse drop in dead time — we measure that at one to three and a half percent of total loss. Their driver needs one resistor; ours needs a clamp device, a negative supply and twenty LUTs. For a converter that never leaves one operating point, theirs may well be the right engineering.

One more thing we have to say plainly. Their netlist is not published, so this is our implementation of their described scheme, in our testbench. The ratio is against that reimplement. It is not a claim against their measured result.

24 · Next Steps — 30 s (14:10)

What is left is bench work, not more simulation.

Place-and-route on a chosen board. Synthesis is done; it stops there because the pin constraints are placeholders, and doing it properly means driving the two hundred megahertz clock from a clock manager rather than off a pin.

And a hardware half-bridge, measured. That is the whole of the remaining risk, and we will say it plainly: everything in this deck is a simulation of a converter that has not been built, resting on one behavioural GaN model.

25–26 · References and close — 15 s (14:25)

These are the papers this design engages with directly; number ten is the base paper, and it is the one we reimplemented and measured against.

Everything in this deck — the netlists, the models, the scripts, the figures — runs on request from the repository on the last slide. Thank you.

If she asks

Short answers. Say the sentence, stop, and let her ask the next one.

"So what is actually your contribution?"

Three blocks the base paper does not have — a digital control word, an always-on Miller clamp, and a selectable negative off rail — and a measured answer to a question the field states as one number: what re-tuning is worth on its own.

"And what is that answer?"

Choosing a better fixed word is worth **26.5 %**. Adapting it per operating point adds **2.6 %** — about **8.9 %** of the total gain. The absolute ceiling on any scheduling scheme is **3.5 %**. So the sensor, the ADC and the lookup table are buying the small share, and that is a design decision we can now make on evidence.

"Why should I believe the numbers?"

Every one of them is regenerated by a named script, the summary file says which script produces which number, and a checker fails the build if a slide and the data disagree. The command on slide twenty re-derives two of them in ten seconds.

"Is the clamp your idea?"

No, and we say so on the slide. Miller clamps and negative off-bias are established practice and we cite four papers. What is ours is that the base paper has neither, and that we measure what each is worth here.

"Have you built it?"

No. It is simulated, in ngspice, on one behavioural GaN model written from the EPC2010C datasheet. That is the largest limitation in the project, it is on slide twenty-four, and a measured half-bridge is the next step.

"Why is your turn-on energy worse?"

Because GaN has no body diode. During dead time the device conducts in reverse, and that costs the threshold plus whatever off-bias you applied. The negative rail that buys us crosstalk margin is paid for there. We measure it at one to three and a half percent of total loss.

"Why ngspice and not a commercial tool?"

Everything here is open source, so there is nothing in the project anyone needs a licence to reproduce. We cross-checked the same circuit in LTspice.

"How does the FPGA fit in?"

The controller is written, verified and synthesised, and its output drives the SPICE power stage directly. On the real part it is twenty LUTs and twenty flip-flops. It is not placed and routed — that is on the next-steps slide.

Before you walk in

- ☐ Slide 1 signed by Dr. Bindu and scanned back in as slide 1.
 - ☐ `bash proof/PREFLIGHT.sh` run on the presentation laptop, green.
- It rehearses both live commands and prints READY at the end.

- ☒ Terminal open in the repository, font size up, slide 20 rehearsed once.
- ☐ Videos on slides 17, 18, 19 and 20 play — open the .pptx, not the PDF.
- ☐ The PDF on a pen drive as the fallback.
- ☐ One rehearsal against a clock. Aim to finish at 14:00, not 14:25.

